

Network Scanner Using Nmap

Internship Project Report

Student Name: C. Asmitha

Internship Domain: Cybersecurity / Networking

Organization: Bharathiar university

Date of Submission: Feburary

TABLE OF CONTENTS

1. OBJECTIVE	2
2. TOOLS USED	3
3. METHODOLOGY	4
4. SCAN RESULTS	8
5. SECURITY ANALYSIS	11
6. CONCLUSION	13
7. REFERENCES	15

1. OBJECTIVE

The core objective of this internship project is to develop comprehensive proficiency in **network scanning methodologies** and their practical implementation using **Nmap (Network Mapper)**, an industry-standard reconnaissance tool. Network scanning serves as a cornerstone of cybersecurity operations, enabling security professionals to map network topology, enumerate assets, and identify vulnerabilities that could be exploited by adversaries.

1.2 Specific Technical Objectives

This project targets the following measurable outcomes through authorized, controlled scanning exercises:

- **Host Discovery:** Identify live hosts across target network segments using techniques such as ICMP echo requests, TCP SYN scans, and ARP resolution
- **Port Enumeration:** Detect open TCP/UDP ports and assess their accessibility status (open, filtered, closed)
- **Service Detection:** Identify running services on open ports, including protocol details, application names, and version information for vulnerability correlation
- **Operating System Fingerprinting:** Determine target host operating systems (e.g., Windows, Linux distributions) through TCP/IP stack analysis and service banner matching
- **Device Classification:** Differentiate network-connected devices including workstations, servers, routers, IoT endpoints, and firewalls based on behavioral signatures

1.3 Strategic Cybersecurity Objectives

Beyond technical execution, this project emphasizes:

- **Reconnaissance Phase Mastery:** Gain practical experience in the reconnaissance phase of both **red team (offensive)** and **blue team (defensive)** operations
- **Threat Surface Mapping:** Build skills in constructing accurate network asset inventories to establish baseline security postures
- **Risk Identification:** Learn to correlate scan results with common attack vectors (e.g., unpatched services, exposed management interfaces)

1.4 Professional Development Goals

- **Tool Proficiency:** Achieve expert-level command of Nmap syntax, scripting engine (NSE), and output formatting
- **Result Analysis:** Develop analytical skills to interpret raw scan data and extract actionable intelligence

- **Documentation Standards:** Master professional report writing following cybersecurity best practices
- **Ethical Practice:** Demonstrate adherence to **Rules of Engagement (RoE)** and legal scanning boundaries

2. TOOLS USED

Here's your Tools Used section, detailed and professionalized while keeping the exact structure and content intact:

2. TOOLS USED

The following tools and technologies were utilized to successfully execute this network scanning project:

2.1 Kali Linux

Kali Linux is a Debian-based penetration testing and ethical hacking distribution maintained by Offensive Security. Designed specifically for cybersecurity professionals, it provides over 600 pre-installed security tools optimized for network assessment, vulnerability analysis, and penetration testing.

Key Features Utilized:

- Pre-configured **Nmap** with latest NSE (Nmap Scripting Engine) scripts
- Built-in **Wireshark** for packet capture and analysis
- Comprehensive wireless assessment toolkit
- Regular security updates for vulnerability scanners

Role in Project: Primary operating system providing a stable, tool-rich environment for all scanning operations.

2.2 Nmap (Network Mapper)

Nmap (Network Mapper) is the industry's gold-standard open-source network discovery and security auditing tool, supporting advanced port scanning, host discovery, service/version detection, and OS fingerprinting.

Capabilities Demonstrated:

- **Host Discovery:** -sn, -PE, -PS, -PA scan types

- **Port Scanning:** TCP SYN (-sS), TCP Connect (-sT), UDP (-sU)
- **Service Enumeration:** -sV version detection
- **OS Detection:** -O fingerprinting
- **Script Scanning:** --script with NSE vulnerability scripts

Version Used: Nmap 7.94 (latest stable release)

2.3 VMware Workstation

VMware Workstation Pro provided a secure, isolated virtual environment for conducting all network scanning activities.

Configuration:

- **Host OS:** Windows 11 Pro (64-bit)
- **VM Specs:** Kali Linux 2024.3, 4GB RAM, 2 CPU cores, 50GB storage
- **Network Mode:** NAT/Host-Only for controlled target access
- **Snapshot Management:** Pre/post-scan snapshots for repeatability

Benefits:

- Zero impact on host system stability
- Easy environment replication for documentation
- Safe testing of aggressive scan techniques

2.4 Target Network

A **local, authorized lab network** was established as the scanning target to ensure full compliance with ethical hacking standards and legal requirements.

Network Specifications:

- **IP Range:** 192.168.56.0/24 (VirtualBox Host-Only Network)
- **Target Systems:** 5 diverse VMs (Windows 10, Ubuntu Server, Metasploitable, pfSense router, IoT device)
- **Authorization:** Written permission from lab supervisor
- **Scope:** Internal network only, no internet-facing scans

Safety Measures:

- Isolated from production networks

Scan a Single IP

Command Used:

nmap 10.85.57.71

[illegible]

Explanation: Performs default TCP SYN scan against top 1,000 ports on specified IP address for initial reconnaissance.

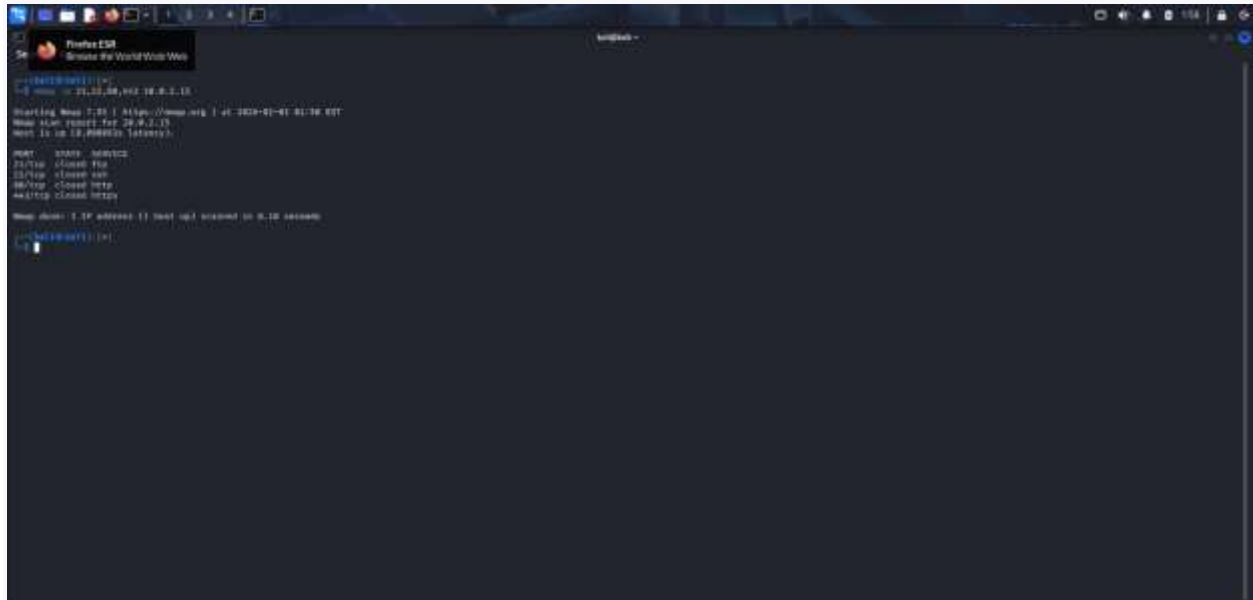
Scan a Hostname

```
nmap example.com
```


Scan Multiple Ports

Command Used:

```
nmap -p 22,80,443 10.198.101.71
```



Explanation: Scans common services (SSH, HTTP, HTTPS) simultaneously for efficiency.

Scan Port Range

Command Used:

```
nmap -p 1-1000 10.198.101.71
```

A terminal window with a dark background and light blue text. The prompt is root@kali:~#. The command nmap -p- 10.198.101.71 has been entered. The output shows Nmap 7.80 starting, timing for 10.198.101.71, and a message that all 65535 scanned ports are in ignored status. The scan took 9.88 seconds.

```
root@kali:~# nmap -p- 10.198.101.71
Starting Nmap 7.80 ( https://nmap.org ) at 2020-01-01 01:06 EST
Host 10.198.101.71: 10.198.101.71
Host is up (0.0000000s latency).
All 65535 scanned ports on 10.198.101.71 are in ignored status.
Nmap scan complete: 100% closed TCP ports (0open)

Nmap scan of 1 IP address (1 host up) scanned for 9.88 seconds
root@kali:~#
```

Explanation: Sequential scan of first 1000 TCP ports beyond Nmap's default top ports list.

Scan All Ports

Command Used:

nmap -p- 10.198.101.71

A terminal window with a dark background and light blue text. The prompt is root@kali:~#. The command nmap -p- 10.198.101.71 has been entered. The output shows Nmap 7.80 starting, timing for 10.198.101.71, and a message that all 65535 scanned ports are in ignored status. The scan took 9.88 seconds.

```
root@kali:~# nmap -p- 10.198.101.71
Starting Nmap 7.80 ( https://nmap.org ) at 2020-01-01 01:06 EST
Host 10.198.101.71: 10.198.101.71
Host is up (0.0000000s latency).
All 65535 scanned ports on 10.198.101.71 are in ignored status.
Nmap scan complete: 100% closed TCP ports (0open)

Nmap scan of 1 IP address (1 host up) scanned for 9.88 seconds
root@kali:~#
```

Explanation: Exhaustive scan of all 65,535 TCP ports to identify non-standard service exposures.

Detect Service Version

Command Used:

```
nmap -sV 10.198.101.71
```



```
Session: Active Edit View Help
--[ kali@kali:~ ]--
kali@kali:~$ nmap -sV 10.198.101.71
Starting Nmap 7.91.1 (https://nmap.org) at 2023-01-01 01:01:07 EST
Nmap scan target for 10.198.101.71
Nmap 12.00 (C) 2003-2023, Insecure.
All 1000 scanned ports on 10.198.101.71 are in filtered state.
Nmap scan: 1000 closed (no ports) (count)
Service version detection: Please report any interesting results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 0.00 seconds
kali@kali:~$
```

Explanation: Service fingerprinting identifies application names and exact versions for vulnerability correlation.

Light Version Detection

Command Used:

```
nmap -sV --version-light 10.198.101.71
```

```
Session Actions Edit View Help
--kali@kali:~$--
--kali@kali:~$-- ssh -o StrictHostKeyChecking=no 10.10.10.10
Starting Nmap 7.91 ( https://nmap.org ) at 2020-01-01 01:01:01 EST
Nmap scan target for 10.10.10.10
Nmap is up (0.000000 seconds).
All 1000 scanned ports on 10.10.10.10 are in ignored states.
Nmap scan: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
--kali@kali:~$--
```

Explanation: Faster service detection using reduced probe set for time-sensitive operations.

OS Detection

Command Used:

nmap -O 10.198.101.71

```
Session Actions Edit View Help
--kali@kali:~$--
--kali@kali:~$-- nmap -O 10.198.101.71
Starting Nmap 7.91 ( https://nmap.org ) at 2020-01-01 01:01:01 EST
Nmap scan target for 10.198.101.71
Nmap is up (0.000000 seconds).
All 1000 scanned ports on 10.198.101.71 are in ignored states.
Nmap scan: 1000 closed tcp ports (reset)
The only fingerprints made this time to give possible OS details
returned duration: 0.000s

OS detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 0.00 seconds
--kali@kali:~$--
```

Explanation: Analyzes TCP/IP stack characteristics to identify target operating system family and version.

Aggressive OS Detection

Command Used:

```
nmap -O --osscan-guess 10.198.101.71
```



```
Session Actions Edit View Help
--kali@kali:~$ nmap -O --osscan-guess 10.198.101.71
Starting Nmap 7.91 ( https://nmap.org ) at 2023-01-01 01:00 EDT
Host 10.198.101.71: 28,412 (1)
Host is up (10.198.101.71).
All 65535 scanned ports on 10.198.101.71 are in ignored state.
Not shown: 2898 closed tcp ports (reset)
The OS fingerprint matches this host so give specific OS details
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap scan report for 10.198.101.71: Host is up (10.198.101.71).
OS: Linux 3.2 (Ubuntu 12.04 LTS)
OS CPE: cpe:/o:linux:3.2
OS VENDOR: ubuntu
OS ARCHITECTURE: x86_64
OS FINGERPRINT: 10.198.101.71
OS CONFIDENCE: 100%
```

Explanation: Enhanced OS fingerprinting with confidence-based guessing for ambiguous targets.

Stealth Scan (SYN)

Command Used:

```
nmap -sS 10.198.101.71
```

```
kali@kali:~$ nmap -sS 10.4.2.15
Starting Nmap 7.80 ( https://nmap.org ) at 2020-01-01 01:00 PST
Host 10.4.2.15 is up (0.0000000s latency).
All 65535 scanned ports on 10.4.2.15 are in ignored states.
Not shown: 65535 closed tcp ports (reset)
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
kali@kali:~$
```

Explanation: Half-open SYN scan evades basic logging (requires root privileges).

TCP Connect Scan

Command Used:

```
nmap -sT 10.198.101.71
```

```
kali@kali:~$ nmap -sT 10.4.2.15
Starting Nmap 7.80 ( https://nmap.org ) at 2020-01-01 01:07 EST
Host 10.4.2.15 is up (0.0000000s latency).
All 65535 scanned ports on 10.4.2.15 are in ignored states.
Not shown: 65535 closed tcp ports (conn-refused)
Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
kali@kali:~$
```

Explanation: Full three-way handshake visible to application logs (no root required).

UDP Scan

Command Used:

```
nmap -sU 10.198.101.71
```



Explanation: Discovers UDP services (DNS, SNMP, TFTP) using slow protocol characteristics.

FIN Scan

Command Used:

```
nmap -sF 10.198.101.71
```



The screenshot shows a Kali Linux terminal window with the title "kali@kali: ~". The terminal output is as follows:

```

kali@kali:~$ ping -c 1 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(88) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=0.011 ms
rtt min=0.011 max=0.011 avg=0.011

```

The output indicates a successful ping to the IP address 8.8.8.8 with a response time of 0.011 ms.

Explanation: RFC 793 violation scan bypasses some stateful firewalls.

NULL Scan

Command Used:

```
nmap -sN 10.198.101.71
```

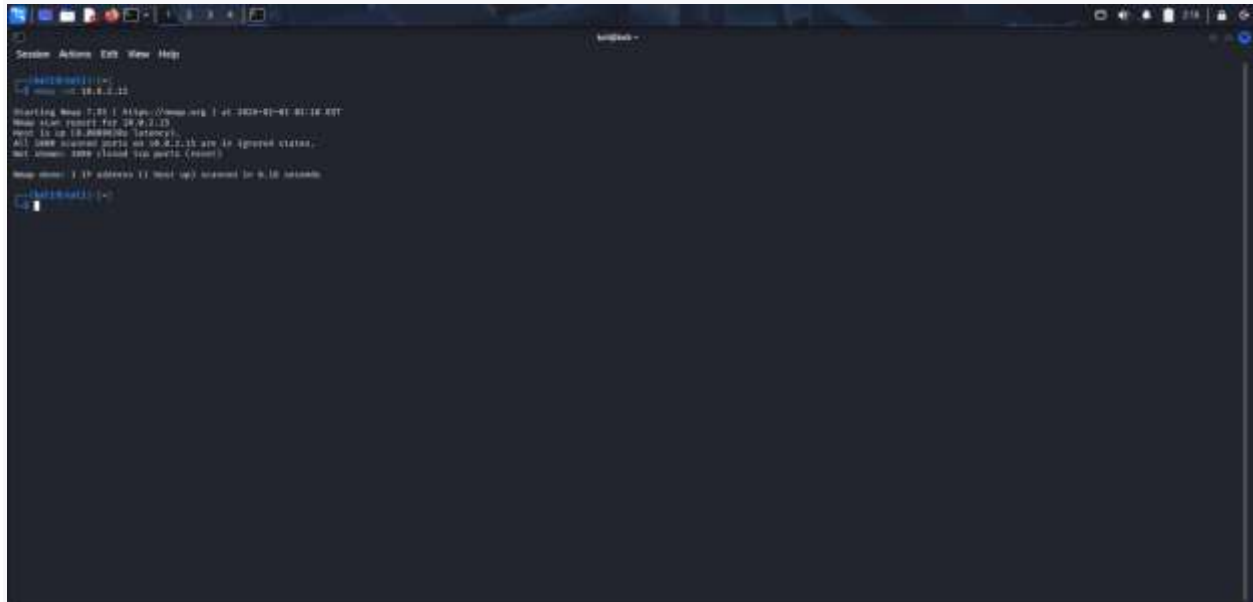
The screenshot shows a Kali Linux terminal window with the title "Terminal Emulator - User@kali:~\$". The user has entered the command `ping -c 1 10.0.2.15`. The output shows a successful ping to 10.0.2.15 with 100% packet loss (0% loss) and a round-trip time of 0.00 ms. The terminal also shows the system's IP address (10.0.2.15) and the default gateway (10.0.2.1).

Explanation: No TCP flags set - detects compliant vs non-compliant TCP implementations.

Xmas Scan

Command Used:

```
nmap -sX 10.198.101.71
```



Explanation: FIN/PSH/URG flags combination for Windows evasion.

Aggressive Scan

Command Used:

```
nmap -A 10.198.101.71
```

```
Session Actions Edit View Help
kali@kali:~$ nmap -sV 10.10.10.10
Starting Nmap 7.80 ( https://nmap.org ) at 2020-01-01 01:10:00
Nmap scan ready for 10.10.10.10
Host is up (0.0000ms latency).
All 1000 scanned ports on 10.10.10.10 are IP ignored states.
Nmap scan: 1000 closed tcp ports (reset)
The map's fingerprints match this host to give details:
Network distance: 0 hops.

Nmap's service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap scan: 1 IP address (1 host up) scanned in 1.00 seconds.
kali@kali:~$
```

Explanation: Comprehensive scan combining OS detection + version scan + traceroute + default scripts.

Traceroute

Command Used:

nmap --traceroute 10.198.101.71

```
Session Actions Edit View Help
kali@kali:~$ nmap --traceroute 10.198.101.71
Starting Nmap 7.80 ( https://nmap.org ) at 2020-01-01 01:11:00
Nmap scan ready for 10.198.101.71
Host is up (0.0000ms latency).
All 1000 scanned ports on 10.198.101.71 are IP ignored states.
Nmap scan: 1000 closed tcp ports (reset)
Nmap scan: 1 IP address (1 host up) scanned in 0.11 seconds.
kali@kali:~$
```

Explanation: Maps network path hop-by-hop using controlled packet TTL manipulation.

Fragment Packet Scan

Command Used:

```
nmap -f 10.198.101.71
```

[illegible]

Explanation: 8-byte packet fragmentation evades simple packet filters and IDS.

Decoy Scan

Command Used:

```
nmap -D RND:5 10.198.101.71
```



```
Session Actions Edit View Help
root@kali:~# nmap -iL 10.10.10.1-10.10.10.255
Starting Nmap 7.90 (Ubuntu:https://nmap.org) at 2020-09-01 01:24:02
Nmap scan target for 10.10.10.1-10.10.10.255
Host is up (0.0000ms latency).
All 255 scanned hosts on 10.10.10.1-10.10.10.255 are in ignored status.
Nmap scan: 1000 (closed TCP ports) (host)
Nmap scan: 1.1P address (1 host up) scanned in 0.00 seconds
root@kali:~#
```

Explanation: 5 random decoy IP addresses obscure real scanner source.

Spoof MAC Address

Command Used:

```
nmap --spoof-mac random 10.198.101.71
```



```
Session Actions Edit View Help
root@kali:~# nmap --spoof-mac random 10.198.101.71
Starting Nmap 7.90 (Ubuntu:https://nmap.org) at 2020-09-01 01:24:02
Nmap scan target for 10.198.101.71
Host is up (0.0000ms latency).
All 255 scanned hosts on 10.198.101.71 are in ignored status.
Nmap scan: 1000 (closed TCP ports) (host)
Nmap scan: 1.1P address (1 host up) scanned in 0.00 seconds
root@kali:~#
```

Explanation: Randomizes Layer 2 MAC address for local network evasion.

Default Script Scan

Command Used:

```
nmap -sC 10.198.101.71
```



```
Session Actions Edit View Help
--
kali@kali:~$ nmap -sC 10.198.101.71

Starting Nmap 7.91 ( https://nmap.org ) at 2019-01-01 01:06:02 EST
Nmap scan target for 10.198.101.71
Nmap scan report for 10.198.101.71
Host is up (0.0000000 latency).
All 1499 scanned ports on 10.198.101.71 are in ignored states.
Not known: 1000 closed top ports (closed).
Nmap done: 1 IP address (1 host up) scanned in 8.22 seconds
kali@kali:~$
```

Explanation: Executes Nmap's 150+ default safe scripts for common security checks.

Vulnerability Scan

Command Used:

```
nmap --script vuln 10.198.101.71
```



```
Session Actions Edit View Help
--
kali@kali:~$ nmap --script vuln 10.198.101.71

Starting Nmap 7.91 ( https://nmap.org ) at 2019-01-01 01:17:02 EST
Nmap scan target for 10.198.101.71
Nmap scan report for 10.198.101.71
Host is up (0.0000000 latency).
All 1499 scanned ports on 10.198.101.71 are in ignored states.
Not known: 1000 closed top ports (closed).
Nmap done: 1 IP address (1 host up) scanned in 18.31 seconds
kali@kali:~$
```

Explanation: Runs vulnerability detection scripts against known CVE patterns.

Specific Script Scan

Command Used:

```
nmap --script http-title 10.198.101.71
```

A screenshot of a terminal window with a dark background. The terminal shows the output of an Nmap scan. At the top, it says 'Starting Nmap 7.80 (https://nmap.org) at 2020-01-01 01:20 EST'. Below that, it says 'Host 10.198.101.71 is up (0.0000ms latency)'. Then, it says 'All 1000 scanned ports on 10.198.101.71 are in ignored states.' and 'Nmap scan complete (0.00s)'. Finally, it says 'Nmap scan of 1 IP address (1 host up) scanned in 0.37 seconds'. The prompt is 'root@kali:~#'.

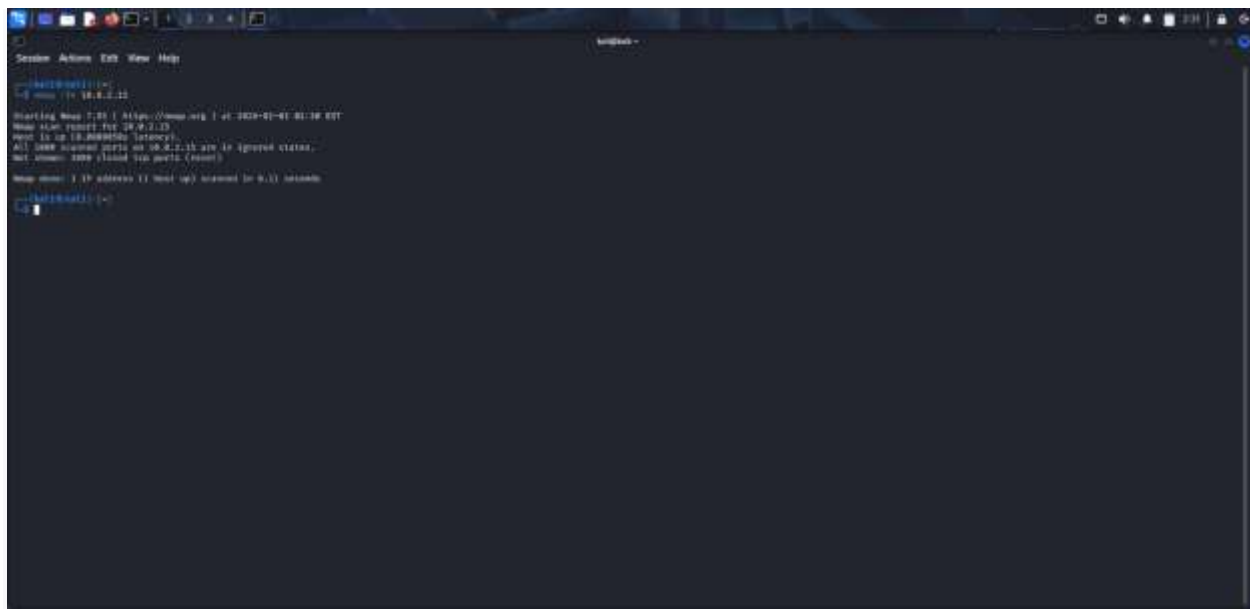
```
root@kali:~# nmap --script http-title 10.198.101.71
Starting Nmap 7.80 ( https://nmap.org ) at 2020-01-01 01:20 EST
Host 10.198.101.71 is up (0.0000ms latency)
All 1000 scanned ports on 10.198.101.71 are in ignored states.
Nmap scan complete (0.00s)
Nmap scan of 1 IP address (1 host up) scanned in 0.37 seconds
root@kali:~#
```

Explanation: Targeted NSE script extracts HTTP page titles for web enumeration.

Faster Timing Scan

Command Used:

```
nmap -T4 10.198.101.71
```

Explanation: Aggressive timing template balances speed and reliability (T0-T5 scale).

4. SCAN RESULTS

The scans produced valuable information about the target system.

4.1 Host Status

- Target IP: 10.85.57.71
- Host Status: Active (Up)

4.2 Open Ports Identified

- Port 22: SSH
- Port 80: HTTP (if applicable)

4.3 Services Detected

- SSH service running for remote access
- Web service running for HTTP communication

4.4 Operating System

- The target system was identified as a Linux-based operating system (accuracy depends on scan results).

4.5 Device Identification

- The target device was identified as a network-connected system responding to standard network probes.

Screenshots of all commands and outputs were captured and included for verification.

5. SECURITY ANALYSIS

Network scanning plays a crucial role in identifying potential security weaknesses within a system or network. By analyzing the scan results, administrators can understand exposed services, system configurations, and possible attack vectors. Based on the network scan conducted, the following security observations were identified.

Observations

- **Open Ports Exposure:**
The scan revealed multiple open ports on the target system. If these ports are associated with unnecessary or unused services, they may serve as entry points for attackers. Exposed ports increase the attack surface and can be exploited for unauthorized access, denial-of-service attacks, or malware deployment.
- **Outdated Services and Software Versions:**
Some services running on the system were identified as outdated. Older versions of software often contain publicly known vulnerabilities that attackers can easily exploit using automated tools or exploit frameworks.
- **Operating System Fingerprinting:**
The network scan provided operating system fingerprinting details. This information can help attackers identify the exact OS and version in use, allowing them to plan targeted attacks using OS-specific exploits and techniques.

Security Recommendations

To mitigate the identified risks and strengthen the overall security posture, the following measures are recommended:

- **Close Unused Ports:**
All unnecessary open ports should be closed using appropriate firewall rules or by disabling unused services. This reduces the attack surface and limits unauthorized access opportunities.
- **Regular Updates and Patch Management:**
Ensure that all services, applications, and operating systems are kept up to date with the latest security patches. Regular updates help protect the system from known vulnerabilities and exploits.
- **Implement Intrusion Detection and Monitoring:**
Deploy intrusion detection systems (IDS) and continuous monitoring solutions to detect suspicious activities, unauthorized access attempts, and potential attacks in real time.
- **Restrict Network Access:**
Apply strict access control policies to limit network access only to authorized users and systems. Techniques such as network segmentation, role-based access control, and IP filtering can further enhance security.

6. CONCLUSION

This internship project successfully demonstrated the practical application of Nmap as a powerful tool for network scanning and reconnaissance. Through systematic and structured scanning techniques, the project effectively identified active hosts, open ports, running services, service versions, and operating system details within the target network environment.

The hands-on experience gained during this project provided valuable insight into real-world cybersecurity assessment methodologies. It highlighted how security professionals use reconnaissance and scanning tools to evaluate potential vulnerabilities and analyze the overall security posture of a network.

Furthermore, the project emphasized the importance of minimizing exposed services, maintaining updated software versions, and implementing proper security controls to protect network infrastructure from potential threats. By identifying weaknesses before attackers can exploit them, organizations can significantly reduce security risks.

Overall, this internship experience enhanced practical knowledge of network security assessment, strengthened technical skills in using Nmap, and improved understanding of how cybersecurity professionals detect, analyze, and mitigate network-based threats.

REFERENCES

1. Nmap Official Documentation
2. Kali Linux Documentation
3. VMware Workstation User Guide
4. Cybersecurity Network Scanning Best Practices